

1

Get your station ready

How do three different liquids change a hydrogel?

Compare vinegar, DI water, and baking-soda solution. Run all three conditions in each of three trials.



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YOUR GROUP NEEDS

4 clear 16 fl oz cups: 3 samples + 1 BASIC STOCK

1 graduated cylinder; 1 stirring spoon per group

Hydrogel powder (9 scoops total); supplied 0.5 g scoop

1 packet of 1.5 g baking soda; DI water; supplied vinegar

pH strips; goggles; timer; marker; paper towels; data sheet

Three conditions. Three trials.

Reuse the same three sample cups after each cleanup. Keep the BASIC STOCK cup for the full lab. Make fresh gel for every trial: nine samples total. Keep all measured amounts the same.

Wear goggles. Keep powder away from your face. Do not taste any material. Tell your instructor about spills.

2

Prepare and label

Make BASIC STOCK once for all three trials.

1.5 g

baking soda

+

150 mL

DI water

1. Label the fourth cup BASIC STOCK. Add the 1.5 g packet.
2. Measure 150 mL DI water with the cylinder and add it.
3. Stir until dissolved using the shared spoon.

Reserve this solution: use 50 mL in each trial.

ACIDIC

NEUTRAL

BASIC

Label the three sample cups ACIDIC, NEUTRAL, and BASIC. Keep the BASIC STOCK cup separate; do not add hydrogel powder to it. Use one shared stirring spoon per group.

Check pH before adding solutions to the gel.

Use a fresh pH strip for each test liquid, follow the strip instructions, and record the reading. Read the actual pH; do not assume a number from the label.

DI = deionized water. Use the supplied water and vinegar. Rinse the spoon and measuring cylinder with DI water and drain before changing liquids or cups.

3

Make three matching gels

Use this recipe in EACH of the three sample cups.

ACIDIC**0.50 g**

hydrogel powder

+ 40 mL

DI water

NEUTRAL**0.50 g**

hydrogel powder

+ 40 mL

DI water

BASIC**0.50 g**

hydrogel powder

+ 40 mL

DI water

1. Add one level scoop of hydrogel powder to each sample cup using the supplied 0.5 g scoop. Keep the scoop dry.
2. Use the graduated cylinder to add 40 mL DI water to each sample cup. Read the bottom of the water curve at eye level. Do not fill the cups.
3. Use the separate spoon to stir gently as each gel forms. Use the same method for all three. Note how the gels look before testing.

Next: add the test liquids.

After preparing all three gels, continue to Step 4. Record any differences; do not add extra powder or water to change how a sample looks.

4

Add, stir, then wait

Match each liquid to its label.

Sample cup	Add to the prepared gel
ACIDIC	50 mL supplied vinegar
NEUTRAL	50 mL DI water
BASIC	50 mL solution from the BASIC STOCK cup

Use the group's one spoon. Before each new cup, wipe any gel clinging to the spoon into the trash, then rinse the spoon with DI water and drain. Rinse and drain the cylinder between different liquids.

10 stirs

same gentle motion

5 minutes

after that sample is stirred

After adding the test liquid, stir that sample 10 times. Start its 5-minute wait as soon as its stirring is complete.

Give every sample the same waiting time.

Use separate timers, write down each start time, or test one condition at a time. Score each sample after its own 5-minute wait.

5**Score, record, repeat**

Use the same 1-5 scale for every sample.

5**Fully solid / no change****4****Slightly smaller / softened****3****Partly liquid****2****Mostly liquid****1****Fully liquid**

Compare with the gel before the test liquid was added. Agree on the closest score, then record a brief observation. Similar scores are valid results.

One trial = all three conditions.

Trial 1: Acidic + Neutral + Basic

Trial 2: three fresh gel samples

Trial 3: three fresh gel samples

Between trials: record first. Scrape gel into the trash can. Wipe visible gel from cups and spoon with paper towels; trash the towels. Rinse in the sink and drain. Reuse the three clean sample cups and repeat Steps 3-5.

6 Your results and cleanup

Group: _____ Date: _____

Condition	pH of test liquid
Acidic	
Neutral	
Basic	

Condition	Trial 1	Trial 2	Trial 3	Mean	SD*
Acidic					
Neutral					
Basic					

Mean = (Trial 1 + Trial 2 + Trial 3) / 3

Example scores 2, 3, 4: mean = 3.0. *Optional sample standard deviation: subtract the mean from each score, square the three differences, add them, divide by 2, then take the square root. For this example, SD = 1.0.

What changed? Which results were most consistent?

Finish: scrape gel into the trash can. Wipe remaining visible gel from cups and spoon into the trash with paper towels. Rinse equipment in the sink. Clean the bench, return equipment, and wash your hands.